Formulaire de trigonométrie

Identités fondamentales

$$\cos^2(x) + \sin^2(x) = 1$$
$$1 + \tan^2(x) = \frac{1}{\cos^2(x)}$$
$$1 + \cot^2(x) = \frac{1}{\sin^2(x)}$$

Formules d'addition

$$\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$$
$$\cos(a-b) = \cos(a)\cos(b) + \sin(a)\sin(b)$$
$$\sin(a+b) = \sin(a)\cos(b) + \cos(a)\sin(b)$$
$$\sin(a-b) = \sin(a)\cos(b) - \cos(a)\sin(b)$$
$$\tan(a+b) = \frac{\tan(a)+\tan(b)}{1-\tan(a)\tan(b)}$$
$$\tan(a-b) = \frac{\tan(a)-\tan(b)}{1+\tan(a)\tan(b)}$$

Formules de duplication

$$\sin(2x) = 2\sin(x)\cos(x)$$
$$\cos(2x) = \cos^2(x) - \sin^2(x) = 2\cos^2(x) - 1$$
$$\tan(2x) = \frac{2\tan(x)}{1-\tan^2(x)}$$

Formules d'angle moitié

$$\sin^2\left(\frac{x}{2}\right) = \frac{1-\cos(x)}{2}$$
$$\cos^2\left(\frac{x}{2}\right) = \frac{1+\cos(x)}{2}$$
$$\tan\left(\frac{x}{2}\right) = \frac{\sin(x)}{1+\cos(x)} = \frac{1-\cos(x)}{\sin(x)}$$

Produit en somme

$$\cos(a)\cos(b) = \frac{1}{2}[\cos(a-b) + \cos(a+b)]$$
$$\sin(a)\sin(b) = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$$
$$\sin(a)\cos(b) = \frac{1}{2}[\sin(a+b) + \sin(a-b)]$$

Somme en produit

$$\cos(a) + \cos(b) = 2\cos\left(\frac{a+b}{2}\right)\cos\left(\frac{a-b}{2}\right)$$
$$\cos(a) - \cos(b) = -2\sin\left(\frac{a+b}{2}\right)\sin\left(\frac{a-b}{2}\right)$$
$$\sin(a) + \sin(b) = 2\sin\left(\frac{a+b}{2}\right)\cos\left(\frac{a-b}{2}\right)$$
$$\sin(a) - \sin(b) = 2\cos\left(\frac{a+b}{2}\right)\sin\left(\frac{a-b}{2}\right)$$

Formules liées aux angles remarquables

$$\cos(\pi - x) = -\cos(x)$$
$$\sin(\pi - x) = \sin(x)$$
$$\tan(\pi - x) = -\tan(x)$$
$$\cos(\pi + x) = -\cos(x)$$
$$\sin(\pi + x) = -\sin(x)$$
$$\tan(\pi + x) = \tan(x)$$
$$\cos\left(\frac{\pi}{2} - x\right) = \sin(x)$$
$$\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$$
$$\tan\left(\frac{\pi}{2} - x\right) = \mathrm{cotan}(x)$$
$$\cos\left(\frac{\pi}{2} + x\right) = -\sin(x)$$
$$\sin\left(\frac{\pi}{2} + x\right) = \cos(x)$$
$$\tan\left(\frac{\pi}{2} + x\right) = -\mathrm{cotan}(x)$$